

New K–Ar age dates of Neogene and Quaternary volcanic rocks from the Ecuadorian Andes: Implications for the relationship between sedimentation, volcanism, and tectonics

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Abstract—New K–Ar ages of Tertiary and Quaternary volcanic and sedimentary sections in the Interandean Depression and the southern Ecuadorian basins give an indication of the duration of sedimentary events and the timing of magmatic and tectonic events in the Ecuadorian Andes. In southern Ecuador, the syn-tectonic infilling of the Cuenca basin began in the early Miocene and ceased by the late Miocene. An andesitic stock in this region shows evidence of late Miocene magmatism. Sporadic development of the Pleistocene volcanic-rich Tarqui Formation suggests, contrary to previously held views, that Plio-Pleistocene volcanism is not as well developed in southern Ecuador. In the Alausí region, an Oligocene age for calc-alkaline volcanism, subsequent to the Macuchi Arc accretion, is confirmed. The existence of younger volcanic events is also proved. In central Ecuador, the Western Cordillera is covered in part by volcanic rocks of the mainly Miocene Pisayambo Formation. The age of the Latacunga Formation of the Latacunga–Riobamba basin (Interandean Depression) is established as late Pliocene to Pleistocene.

Resumen—Este artículo presenta nuevas edades radiocronológicas de los depósitos volcano-sedimentarios terciarios y cuaternarios de las cuencas intramontañas de la Depresión Interandina y del sur del Ecuador. Estas edades permiten precisar la duración de los eventos sedimentológicos, magmáticos y tectónicos de los Andes ecuatorianos. En el sur del Ecuador, el relleno sintectónico de la cuenca de Cuenca empezó durante el Mioceno temprano para detenerse en el Mioceno tardío. Un magmatismo del Mioceno tardío está comprobado por la evidencia de una intrusión andesítica. La existencia esporádica de la Formación volcánica Tarqui, de edad Pleistocena, sugiere que el volcanismo plio-pleistoceno no sea tan bien desarrollado en el sur del Ecuador como se suponía. En la región de Alausí, se comprueba la edad oligocena de un volcanismo calco-alcálico, posterior a la acreción del Arco Macuchi. La existencia de eventos volcánicos más jóvenes está probada. En el centro del Ecuador, la Cordillera occidental está cubierta por la Formación volcánica Pisayambo de edad miocénica. En el centro de la Depresión Interandina, en la cuenca de Latacunga–Riobamba, se establece la edad pliocena tardío a pleistocena de la Formación Latacunga.

INTRODUCTION

THE ACTIVE CONTINENTAL MARGIN of Ecuador is characterized by subduction of the Nazca plate under the South American plate. Major magmatic activity and development of intermontane basins, with terrestrial sedimentation, characterize the geodynamic evolution of the Andes throughout the Neogene. The basins are controlled by ongoing compressive tectonics (Noblet *et al.*, 1988; Lavenu and Noblet, 1989). Ages of many of the volcanic and sedimentary units in these basins are still unknown or imprecise, so correlations are generally made on the basis of facies similarities. Such correlations are of questionable validity. New K–Ar ages are presented here for Tertiary and Quaternary volcanic formations of the In-

terandean Depression and southern Ecuador (Fig. 1). These help to establish a chronology for the tectono-sedimentary evolution of this part of the Andes.

GEODYNAMIC SETTING

Two domains characterize the Ecuadorian Andes: one extending from the Eastern Cordillera to the Amazonian Lowlands and belonging to the South American continent, the other made up of two accreted terranes and corresponding to the Western Cordillera and the Coastal Plain. The accreted terranes are, in the south, the Amotape-Tahuin terrane (of continental origin) that was accreted during the Early Cretaceous (Mourier *et al.*, 1988), and in the north, the Piñon-Macuchi terrane, composed of a basaltic oceanic floor and an oceanic volcanic arc, that was accreted during Late Cretaceous to Eocene times (Feininger and Bristow, 1980; McCourt *et al.*, 1984; Lebrat, 1985; Lebrat *et al.*, 1985a,b; Bourgois

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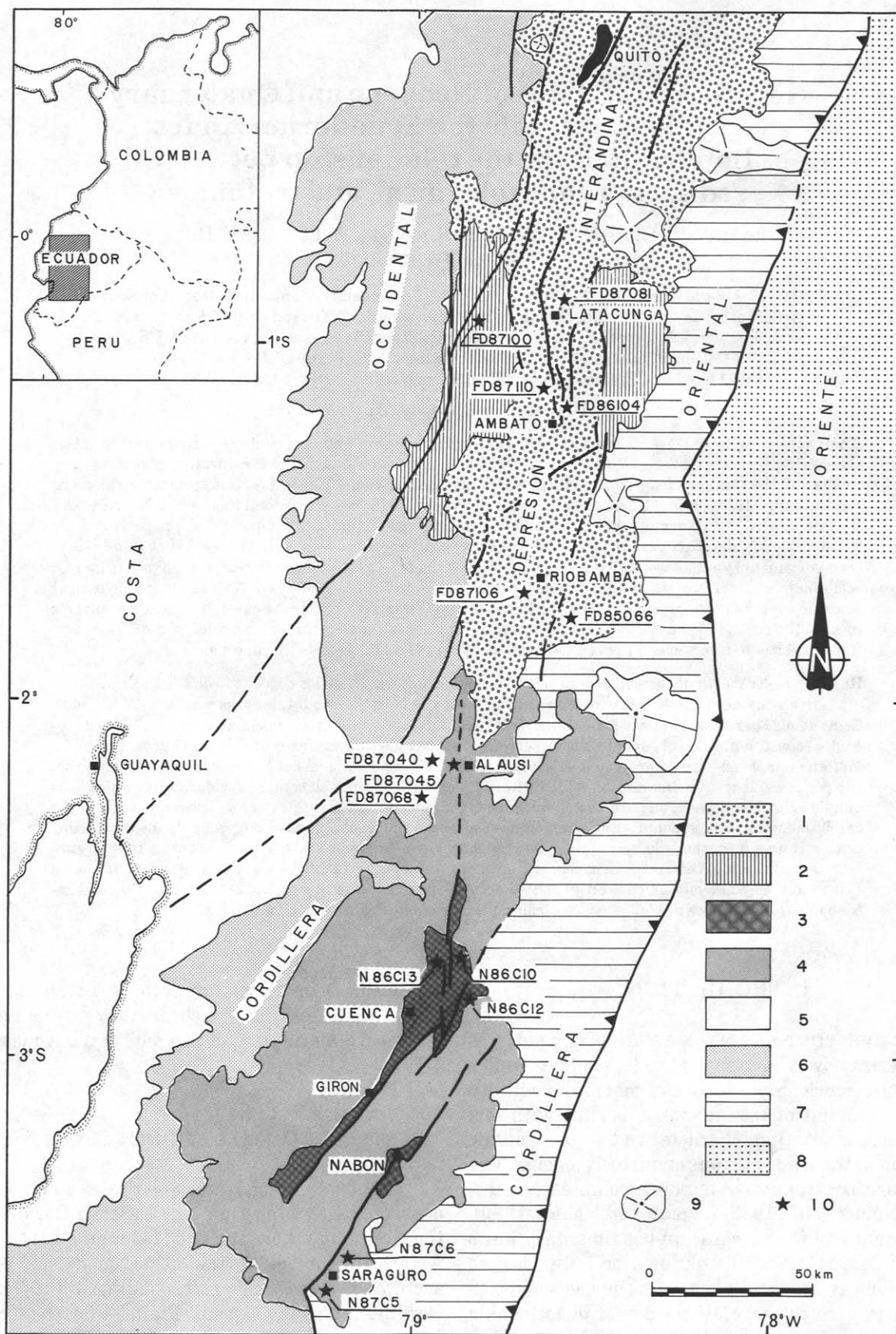


Fig. 1. Simplified morphostructural and geological framework of the central Ecuadorian Andes (from Baldock, 1982): 1, Plio-Quaternary volcaniclastic infilling of the Interandean Depression; 2, Miocene volcaniclastic deposits of the Pisayambo Formation; 3, Tertiary sedimentary infilling of the Cuenca and Nabon basins; 4, undifferentiated Tertiary volcanic rocks; 5, Coastal Plain; 6, Western Cordillera; 7, Eastern Cordillera; 8, Subandean Zone and Oriente; 9, main Quaternary volcanoes; 10, location of studied samples.

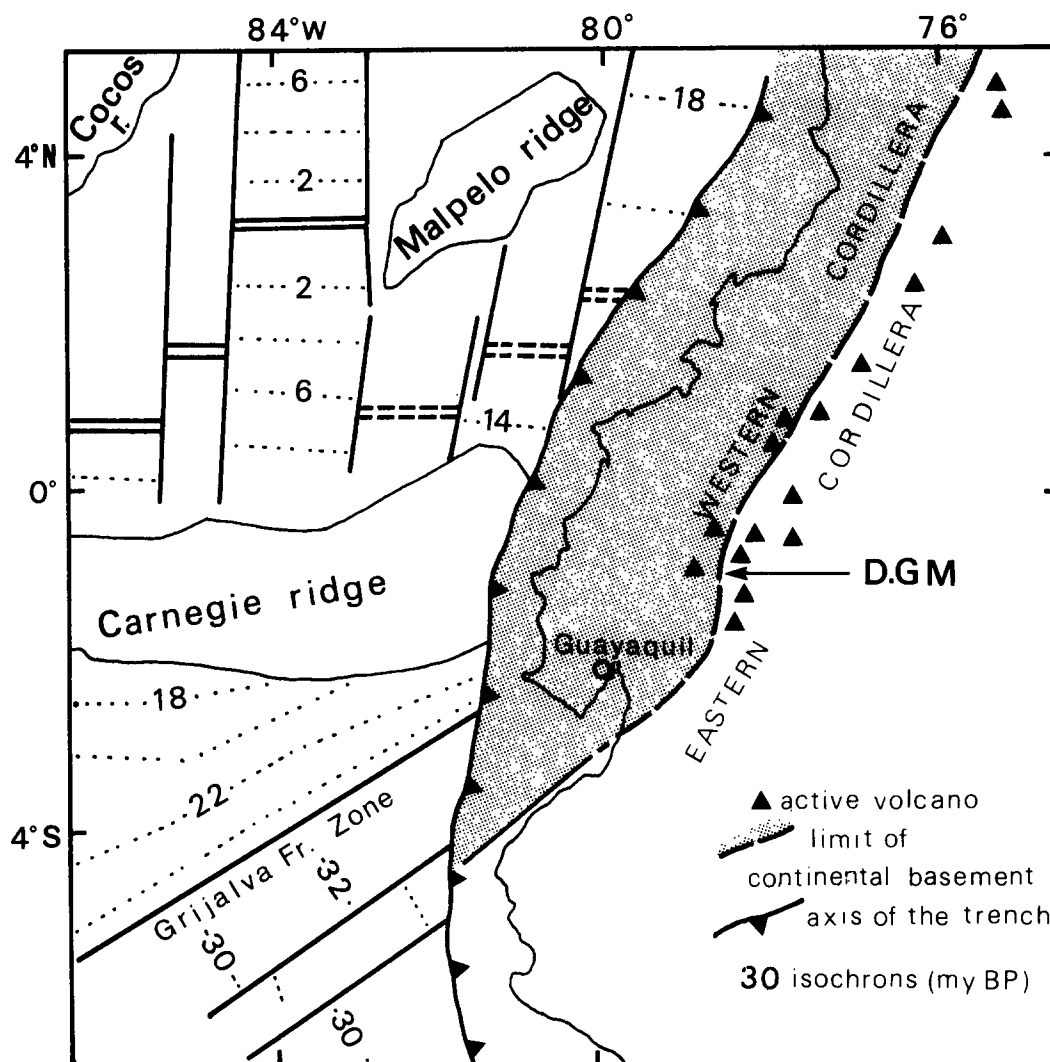


Fig. 2. Simplified sketch map of oceanic and continental areas around the Ecuadorian Trench (from Lonsdale, 1978); DGM, Dolores Guayaquil Megashear.

et al., 1985; Egüez, 1986; Lebrat and Mégard, 1986; Mégard and Lebrat, 1986; Roperch *et al.*, 1987). The suture between these terranes and the South American continent is clearly recognized only in the north, where it is defined by the Dolores-Guayaquil Megashear (Fig. 2) (Case *et al.*, 1971, 1973; Campbell, 1974a,b; Meissner *et al.*, 1976; Feininger and Seguin, 1983; Moberly *et al.*, 1982).

During the Oligocene, subduction of the Farallon plate under South America was responsible for the major uplift stage of the whole chain. This period corresponds to the Huigra-Tandapi volcanic arc of Egüez *et al.* (1988), which developed after the accretionary stage of the Macuchi oceanic volcanic arc and before the Miocene-Recent continental volcanic arc. Subsequent Oligocene volcano-sedimentary nonmarine deposition is represented by the deposits of the Saraguro Group, in the south (Baldock, 1982), and their probable equivalents to the north (Egüez *et al.*, 1988; Van Thournout *et al.*, 1990).

At about 25 Ma, the Farallon plate broke up into the Cocos and Nazca plates (Herron, 1972; Hand-

schumacher, 1976; Hey, 1977; Hey *et al.*, 1977; Minster and Jordan, 1978; Wortel and Cloetingh, 1981; Duncan and Hargraves, 1984). This event may have triggered the opening of the Interandean basins (Noblet *et al.*, 1988) during the Miocene-Recent volcanic arc activity. Aligned along major regional faults and filled by nonmarine sediments and volcanic deposits, the basins are, from north to south, the Chota, Latacunga-Riobamba, Cuenca, Nabon, Loja, Vilcabamba, and Zumba.

Since the early Pliocene, two different oceanic floors, separated by the Grijalva Fracture Zone, have been subducting in the Ecuadorian trench. In the northern part, the young oceanic plate (<20 Ma) produced by the Cocos-Nazca spreading center (Galapagos Rift Zone) is being subducted at an angle of 25-35° (Rea and Malfait, 1974; Lonsdale, 1978; Lonsdale and Klitgord, 1978); in the southern part, the older oceanic crust (>30 Ma) is being subducted at an angle of only 15°. This low angle of subduction may explain the gap in volcanic activity in the southern Ecuadorian Andes (Hall and Wood, 1985; Barberi *et al.*, 1988).

GEOLOGIC SETTING

In southern Ecuador, the Cenozoic formations of the Cuenca basin (Bristow, 1973) consist of four main lithostratigraphic groups (Noblet *et al.*, 1988).

The first group, representing the youngest part of the basement of the basin, consists of the extensive volcanic Saraguro Formation (Kennerley, 1973; Bristow and Hoffstetter, 1977) — a part of the Saraguro Group (Baldock, 1982). Baldock included in this Oligo-Miocene group other dispersed volcanic deposits, some of them doubtfully dated: the Alausí, Chinchillo, Saraguro, and Loma Blanca Formations. Ages obtained are 21.4 ± 0.8 Ma and 26.8 ± 0.7 Ma (Kennerley, 1980) and 28.9 ± 1.40 Ma (Barberi *et al.*, 1988) from the Saraguro Formation, indicating a late Oligocene age, and 14.2 and 19.5 Ma from the Chinchillo Formation (Kennerley, 1980). In the Cuenca area, large areas of volcanoclastic rocks mapped as the Tarqui Formation, which is of Pleistocene age, may in fact correlate with the Saraguro Formation (Noblet *et al.*, 1988; Egüez and Noblet, 1988).

The second group, which corresponds to the detrital Biblian Formation (Sheppard, 1934), unconformably overlies the Saraguro Formation (Egüez and Noblet, 1988; Noblet *et al.*, 1988) and older rocks (Baldock, 1982). Its age is uncertain. According to Bristow (1973) and Madden *et al.* (1989), it is Miocene (middle Miocene, according to Carlini *et al.*, 1989). A notoungulate (Toxodontidae) found by Repetto (1977) indicates a middle Miocene age; however, the location Repetto gave corresponds to the location of the younger Mangan Formation on government maps. Fossils collected by Bristow and Parodiz (1982) indicate an early Miocene age. Andesitic rocks dated at 19.7 ± 0.5 Ma and 21 ± 0.6 Ma were considered by Kennerley (1980) as coeval with the top of the Biblian Formation, but the sample coordinates correspond to the Descanso andesitic stock, which is not in contact with the Biblian Formation (Egüez and Noblet, 1988).

The third lithostratigraphic group consists of the Loyola, Azogues, and Mangan Formations, which unconformably overlie the Biblian Formation. Noblet *et al.* (1988) correlated the Guapan Formation with the Azogues Formation (following Bristow, 1973) and the Santa Rosa and the Turi Formations with the rather younger Mangan Formation. According to sedimentologic observations, the Guapan, Santa Rosa, and Turi 'formations' correspond to lateral facies variations within the Azogues and Mangan Formations. Noblet *et al.* (1988) considered them as members of these formations. On the basis of fossil evidence, the Loyola Formation (Marshall and Bowles, 1932; Liddle and Palmer, 1941; Parodiz, 1969; Bristow, 1973; Roberts, 1975; Bristow and Parodiz, 1982) and the Azogues Formation (Bristow, 1973) could belong to the lower to middle Miocene. The Mangan Formation is undifferentiated Miocene.

The youngest group unconformably overlies the others. It consists of volcano-detrital deposits of the

Llacao and Tarqui Formations, both regarded as Pleistocene (Bristow, 1973).

North of the Cuenca basin, Tertiary deposits of the Alausí region consist of volcanic rocks of the Alausí Formation, according to the government maps. This formation is considered as Paleogene (Oligocene) and coeval with the Saraguro Formation (Bristow and Hoffstetter, 1977) and was included in the Saraguro Group by Baldock (1982). It unconformably overlies the metasedimentary and volcanic upper Mesozoic Paute Group and the volcanic upper Mesozoic Macuchi Formation (Baldock, 1982). In this area, other volcanic and sedimentary deposits are represented by the Pleistocene Tarqui and Palmira Formations (Baldock, 1982).

In the Interandean Depression, as well as along the Western and Eastern Cordilleran margins, sedimentary and volcanic rocks are considered, without any fossil evidence or isotopic data, as Neogene to Quaternary in age (Baldock, 1982): the Moraspamba Formation is considered as Miocene; the Pisayambo Formation, unconformably overlying the Moraspamba Formation, is regarded as late Miocene to Pliocene (Baldock, 1982) or Pleistocene; the volcanic Sicalpa Group is regarded as Pliocene; finally, the Altar Group and the Latacunga Group could be Pleistocene. The latest Quaternary volcanic deposits consist of the Chalupas flow (Beate, 1985; Barberi *et al.*, 1988) and the Cangagua deposit, which overlies portions of the previous units.

ANALYSES AND RESULTS

New K-Ar data, obtained in 1989, are presented in Table 1. Procedures used for sample preparation, potassium determination, argon extraction and purification, and mass isotopic analysis were slightly modified from those of Bonhomme *et al.* (1975). The decay constants used were those recommended by Steiger and Jäger (1977). Error was calculated following Mahood and Drake (1982). Argon calibration mainly used the G1-0 standard sample (Odin, 1982), with a value of 24.82 ± 0.14 nl/g (1 σ). In the Laboratoire de Géologie et Minéralogie of the Institut Dolomieu (Grenoble, France), the average for this sample in 1989 was 24.86 ± 0.19 nl/g (1 σ). Other standard minerals give results within $\pm 2\%$, 1 σ of recommended values. For instance, HDB1 biotite gives 7.625 ± 0.065 , 1 σ , and LPG6 biotite gives 44.05 ± 0.36 , 1 σ , compared with 7.7 nl/g (H. J. Lippolt, pers. commun., 1989) and 43.10 nl/g (Odin, 1982), respectively.

Some of the samples studied showed low radiogenic argon content. The effect of this on the reliability of the dating is discussed below. In the Andes, particularly in Ecuador, most formations were subjected to subsequent thermo-tectonic events, which may have caused total or partial loss of radiogenic argon. To try to assess this, we used at least two phases of each sample analyzed whenever pos-

Table 1. New K–Ar analytical data and age determinations on volcanic rocks from southern and central Ecuador.

Sample	Geologic Formation	Location		Petrographic Definition	Analyzed Fraction	K ₂ O (%)	⁴⁰ Ar _{rad}		Age (Ma ± 1σ)
		Latitude S	Longitude W				⁴⁰ Ar _{tot} (%)	⁴⁰ Ar (nl/g)	
N87C5	Saraguro	2°50'	78°54'	Andesite	PLG	1.37	76.3	1.576	35.3 ± 0.9
N86C12	Biblian	2°42'	78°53'	Rhyolitic tuff	WR	2.36	71.3	1.892	24.7 ± 0.6
					PLG	0.53	70.9	0.378	22.0 ± 0.8
N86C13	Mangan	2°44'	78°54'	Rhyolitic tuff	PLG	0.58	70.1	0.306	16.3 ± 0.7
N86C10	Cojitambo	2°45'	78°53'	Andesite	PLG	1.28	66.6	0.292	7.1 ± 0.3
N87C6	Pisayambo	3°39'	79°15'	Andesite	PLG	1.16	71.6	0.306	8.2 ± 0.4
FD87040	Saraguro	2°08'	78°57'	Andesite	WR	0.90	38.7	0.613	21.0 ± 1.0
					PLG	1.04	34.6	0.919	27.2 ± 0.9
FD87068	Saraguro	2°17'	78°59'	Andesite	WR	0.95	70.2	1.112	35.9 ± 0.9
					PLG	0.30	58.5	0.348	35.5 ± 1.3
FD87045	Pisayambo	2°10'	78°51'	Andesite	WR	1.39	66.2	0.356	7.9 ± 0.4
					PLG	0.44	48.0	0.126	8.8 ± 0.4
					—		58.5	0.178	12.5 ± 0.9
FD87100	Pisayambo	0°58'	78°52'	Andesite	WR	1.12	49.3	0.331	10.0 ± 1.3
					PLG	0.33	32.7	0.107	9.1 ± 0.5
FD87106A	Sicalpa	1°44'	78°41'	Acidic tuff	WR	1.84	88.9	0.214	3.59 ± 0.28
					PLG	0.36	37.8	0.060	5.14 ± 1.11
					—		20.2	0.036	3.12 ± 0.39
FD87106B	Sicalpa	1°44'	78°41'	Acidic tuff	WR	1.86	45.1	0.159	2.65 ± 0.21
					PLG	0.39	48.6	0.060	4.76 ± 0.57
FD85066	Altar	1°47'	78°36'	Andesite	WR	1.97	46.7	0.231	3.53 ± 0.94
FD86104	Latacunga	1°11'	78°34'	Andesite	PLG	1.07	28.5	0.060	1.73 ± 0.35
FD87081	Latacunga	0°55'	78°35'	Basaltic andesite	WR	1.82	22.1	0.109	1.85 ± 0.19
FD87110	Latacunga	1°09'	78°38'	Andesite	WR	1.21	32.0	0.055	1.40 ± 0.29

WR, whole rock; PLG, plagioclase

sible. The plagioclase of volcanic rocks is generally a reliable chronometer unless subjected to thermal resetting, but volcanic glass (if present) is easily reset by devitrification. Glass-free lavas are generally reliable chronometers. Tuffs present special problems: if present, glass may loose argon by devitrification; the ash fraction, which is particularly susceptible to alteration, will also be subject to argon loss; the presence of a detrital fraction may generate an age that pre-dates eruption; tuffs are often reworked and may thus give ages greater than that of deposition. These factors must be borne in mind when assessing K–Ar ages on tuffs.

The low argon content of young rocks and minerals may present special problems, particularly when only a limited quantity of sample is available for analysis. To assess this problem, we have made replicate analyses on the MDD 70 phonolite prepared by J. C. Baudron of the BRGM (France), which yields *ca.* 0.040 nl rgAr/gm. Our analyses give 0.048 ± 0.020 nl/g (1σ). We consider it prudent to expect a comparable error of *ca.* ±40% for all samples containing less than 0.2 nl rgAr/gm. This is supported by the duplicate analyses on plagioclase samples FD7045 and FD87106. It will be observed that the calculated errors are considerably less than the errors indicated by the replicate analyses. The conventional method of assessing error propagation, particularly at low radiogenic argon levels and high atmospheric argon contamination, has long been recognized as inadequate (see, for example, Mahood and Drake, 1982). The geologic significance of the

ages of samples containing less than about 0.2 nl/gm radiogenic argon must thus be assessed with great caution.

STRATIGRAPHIC IMPLICATIONS

The new data reported here allow more coherent stratigraphic correlations between the different areas. Some are at variance with previously held views, as expressed in the published maps (Fig. 3). Changes in correlation supported by the data are discussed here on an area-by-area basis.

Saraguro Area

Several observations can be made concerning the age of the Saraguro Formation in southern Ecuador:

- 1) Sample N87C5, an andesite collected 5 km south of Saraguro, gave an age of 35.3±0.9 Ma, corresponding to the early Oligocene.
- 2) The age of 26.8 Ma reported by Kennerley (1980) corresponds to the late Oligocene. According to the sampling coordinates, our field observations show that this rhyolite corresponds to the top of the Saraguro Formation.
- 3) The age of 21.4 Ma reported by Kennerley (1980) on a rhyolitic lava is questionable. The sample was collected near the major regional Girón fault

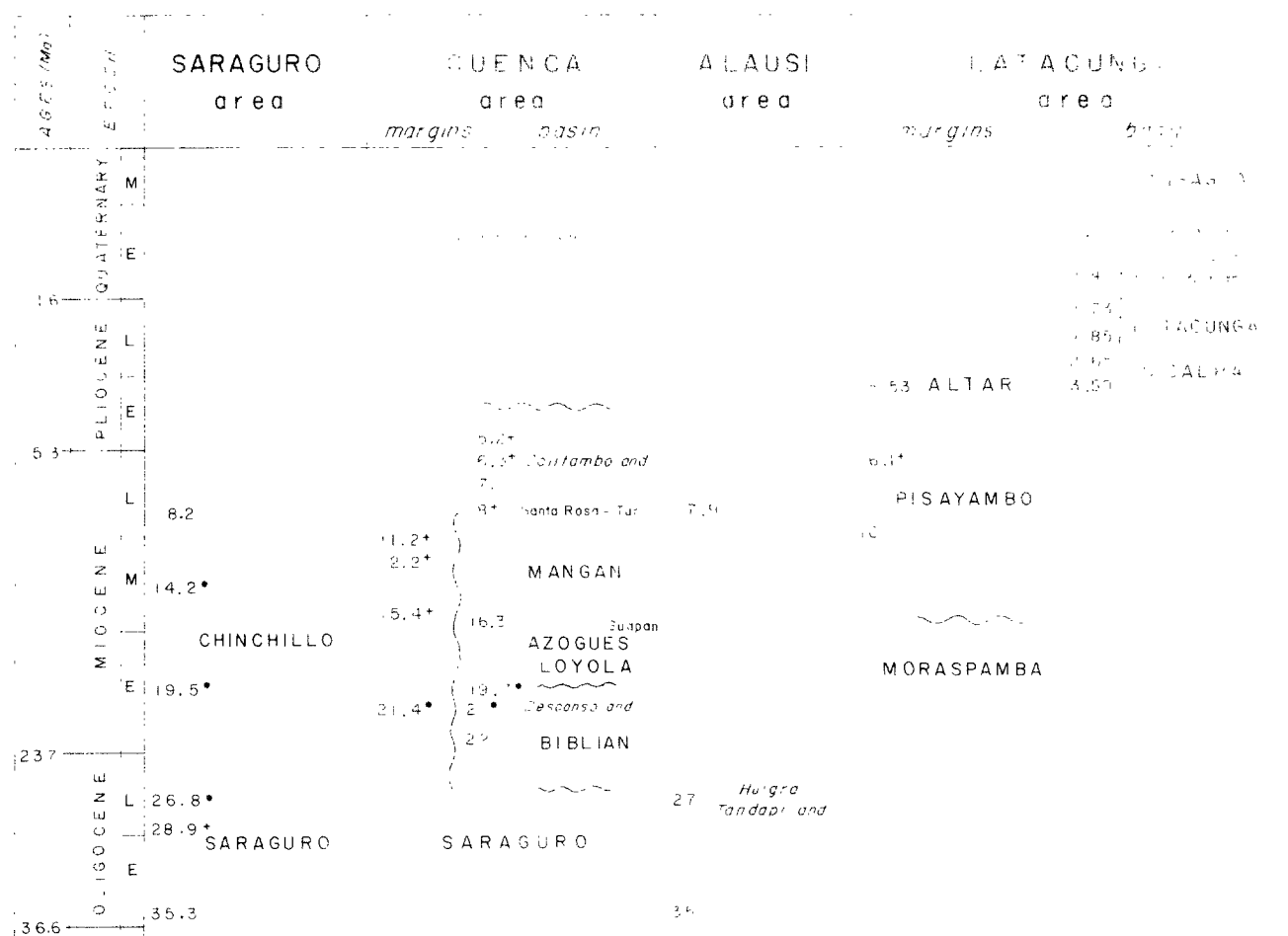


Fig. 3. Chronological correlations between Cenozoic formations: *, radiometric ages of Kennerley (1980); +, radiometric ages of Barberi *et al.* (1988).

system, which controlled the Miocene sedimentation and volcanism of the Cuenca and Girón basins (Noblet *et al.*, 1988; Noblet and Marocco, 1989). Thus, this age might be related more to the syn-sedimentary Miocene volcanism than to the volcanic basement of the basins. Indeed, in the Cuenca basin (see below) the dated 22 Ma Biblian Formation overlies the Saraguro Formation.

In our opinion, the age of the Saraguro Formation is better defined by the 35.3 Ma and 26.8 Ma ages than the 35.3–21.4 Ma interval. The formation thus corresponds to most of the Oligocene Epoch.

In the Saraguro area, the Chinchillo andesitic formation, which overlies the Saraguro Formation, is dated by Kennerley (1980) at 14.2 ± 0.5 Ma (andesite porphyry) and 19.5 ± 0.4 Ma (dike), placing the age of the formation close to the limit between the early and middle Miocene.

Cuenca Area

In the Cuenca basin, 1 km east of the village of Biblian, a tuff sample (N86C12) was collected from the dominantly sedimentary (red beds) Biblian For-

mation, which rests on Cretaceous and Oligocene 'basement.' Dating gave two ages: 22.0 ± 0.8 Ma (plagioclase) and 24.7 ± 0.6 Ma (whole rock). The more reliable age is probably 22 Ma (early Miocene), which makes the formation chronologically older than the Descanso andesite. Sample N86C13, a tuff collected from the lower part of the Mangan Formation, gave a middle Miocene age of 16.3 ± 0.7 Ma, in contrast to the late Miocene age that had been assigned to the formation. The fossil collected by Repetto (1977) also indicated a middle Miocene age. A tuff collected by Barberi *et al.* (1988) south of Cañar, 38 km NNE of Cuenca, gave a late Miocene age of 8.0 ± 0.08 Ma. From geologic maps of the area, this last sample could correspond to the supposedly Pliocene Turi Formation, considered by Noblet *et al.* (1988) to be a member of the top of the Mangan Formation. An unambiguous interpretation of these data is not possible because tuffs are susceptible to alteration that may give overly young ages and the presence of a detrital fraction may give overly old ages.

The Cojitambo andesitic stock (N86C10), which cuts the Biblian and Loyola Formations (light contact metamorphism observed by B. Beate, pers. commun., 1992), gave an age of 7.1 ± 0.3 Ma. Although

older than the ages reported by Barberi *et al.* (1988) (5.2 ± 0.2 Ma and 6.33 ± 0.2 Ma), this confirms late Miocene volcanic activity.

Sample N87C6, an andesite, was collected 5 km NE of Saraguro in an area that geologic maps show as Tarqui Formation. Its age of 8.2 ± 0.4 Ma does not substantiate this interpretation and places the andesite in the late Miocene. Similar results have been obtained by Barberi *et al.* (1988), who reported ages of 11.2 ± 0.3 Ma, 12.2 ± 0.4 Ma, and 15.4 ± 0.7 Ma, corresponding to middle Miocene on volcanic rocks originally assigned to the Tarqui Formation. Thus, the Tarqui Formation does not appear to be as extensive as originally thought. Some of the rocks previously mapped as Tarqui Formation correspond in fact to products of Miocene volcanism, which are well developed in the basins as well as on their margins. However, the existence of a Pleistocene Tarqui Formation could be indicated by the presence of thin volcano-sedimentary deposits unconformably overlying Miocene sediments in the Cuenca basin (Llacao Formation) and by the morphology of some calderas in the western margin of the Cuenca basin (Van Thournout and Guzman, 1988; Perez, 1990).

Alausí Area

Samples FD87040 and FD87068 were collected from andesites shown as the Upper Cretaceous–Paleogene Macuchi Formation on the state map (1982). Sample FD87040 (12.5 km NW of Alausí) gave ages of 27.2 ± 0.9 Ma (plagioclase) and 21.0 ± 1.0 Ma (whole-rock). We consider the older age to be the most reliable, which places this andesite in the late Oligocene. Sample FD87068, collected 20 km SW of Alausí, indicates an early Oligocene age (35.9 ± 0.9 Ma and 35.5 ± 1.3 Ma). Agreement between the whole-rock and plagioclase ages gives confidence in the reliability of these results, which yield a mean age of 35.7 Ma. Thus, these two samples do not belong to the Macuchi Formation but probably to unmapped remnants of the Alausí Formation (Saraguro Group: Baldock, 1982). This volcanic episode is coeval with the Oligocene Saraguro Formation of southern Ecuador (*cf.* N87C5 and N86C12). Alternatively, following Egüez *et al.* (1988), these samples could belong to the Huigra-Tandapi Formation, which is part of the Huigra-Tandapi continental volcanic arc of Oligocene age, overlying the Macuchi arc.

Sample FD87045 was collected 2.5 km WSW of Alausí from andesites shown as the Alausí Formation on government maps. However, this andesite gives ages ranging from 12.5 ± 0.9 Ma to 7.9 ± 0.4 Ma. The latter is considered more reliable, due to the relatively higher K_2O and radiogenic ^{40}Ar contents. The low radiogenic argon content of the plagioclase, close to 0.15 nl/g, does not allow a high level of confidence; at this concentration, replicate analyses indicate an error (1 σ standard deviation) of about $\pm 13\%$. Thus, this rock, which has a late Mio-

cene age, may represent a volcanic episode younger than the Alausí andesite and may possibly belong to a southern extension of the upper Miocene Pisayambo Formation (see below).

Latacunga-Riobamba Area

Volcanic and volcano-sedimentary deposits estimated to be Miocene to Pleistocene in age are widely developed in central Ecuador, in the Western and Eastern Cordilleras and in the Interandean Depression. Our results indicate the ages of various units.

In the Western Cordillera, sample FD87100, collected 11 km SSE of Laguna Quilotoa from the base of the Pisayambo Formation (Baldock, 1982) — at an altitude of 3600 meters and near the unconformity with the folded Morapamba Formation,[‡] gave ages of 9.1 ± 0.5 Ma on plagioclase and 10.0 ± 1.3 Ma on whole rock, indicating a late Miocene age for the formation. A dacite from the basement of the Quilotoa volcano, dated at 6.10 ± 0.60 Ma by Barberi *et al.* (1988), may correspond to the upper part of the Pisayambo Formation.

Samples of the folded volcanic and volcanoclastic Sicalpa Formation, collected 7 km SSW of Riobamba, gave Pliocene ages of 5.14 ± 1.11 Ma, 3.59 ± 0.28 Ma, and 3.12 ± 0.39 Ma from FD87106A and 4.76 ± 0.57 Ma and 2.65 ± 0.21 Ma from FD87106B. In each case, the most reliable result from the purely analytical point of view is that of the whole rock — *i.e.*, 3.59 Ma and 2.65 Ma. Nevertheless, if these tuffaceous samples contain some detrital contribution, the ages may be too old. As discussed above, the small amount of radiogenic argon makes the results obtained on these plagioclases less reliable. Whole-rock analyses place this formation in the late Pliocene. On the border of the Eastern Cordillera, volcanic rocks from near the base of the Altar volcano (Altar Group: Baldock, 1982) were collected from Loma Bellavista, 13 km SSE of Riobamba. They gave a late Pliocene age, 3.53 ± 0.94 Ma (FD85066), and are evidently coeval with the Sicalpa Formation. The later (higher altitude) volcanic deposits of the Altar volcano (5270 m) are still presumed to be of Pleistocene age.

In the Interandean Depression, our recent field studies in the Latacunga-Riobamba zone show four distinct units (Fig. 4):

- 1) A basal volcanoclastic unit, previously assigned to the Pisayambo Formation, consists of lahars, lava flows, volcanic breccias, and occasionally fluvial sediments at the top. These deposits are deformed by syn-sedimentary N/S-trending folds.

[‡]In this part of the Western Cordillera, between latitudes $0^{\circ}40'S$ and $1^{\circ}20'S$, the uplands range from 4000 to 4500 meters in altitude. The Pisayambo Formation overlies the folded older formations (Moraspamba, Macuchi, or Yunguilla) on a horizontal surface at an elevation of 3500–3600 meters. Its present thickness does not exceed 1000 meters.

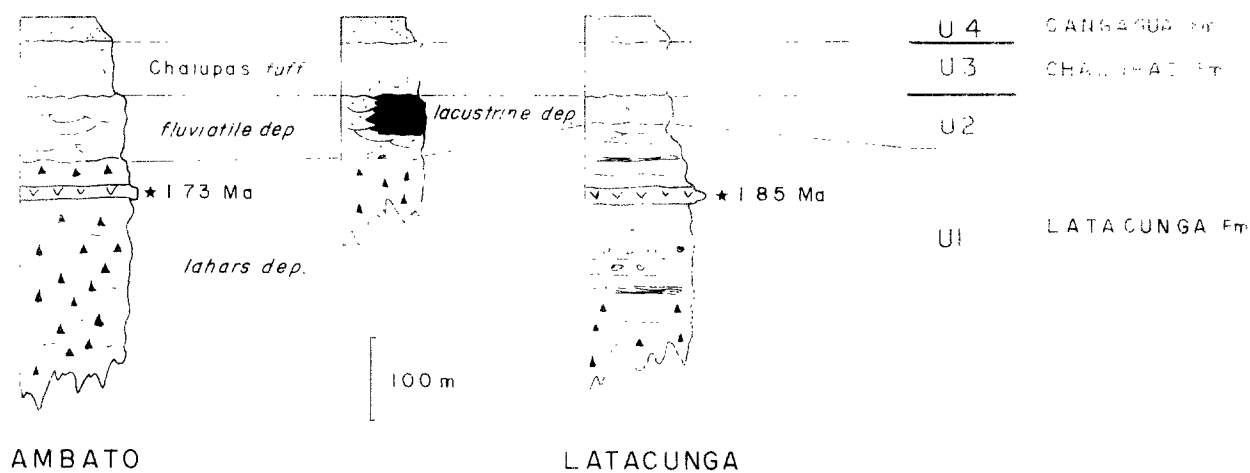


Fig. 4. Cross sections and subdivisions of the Plio-Pleistocene deposits in the Latacunga area.

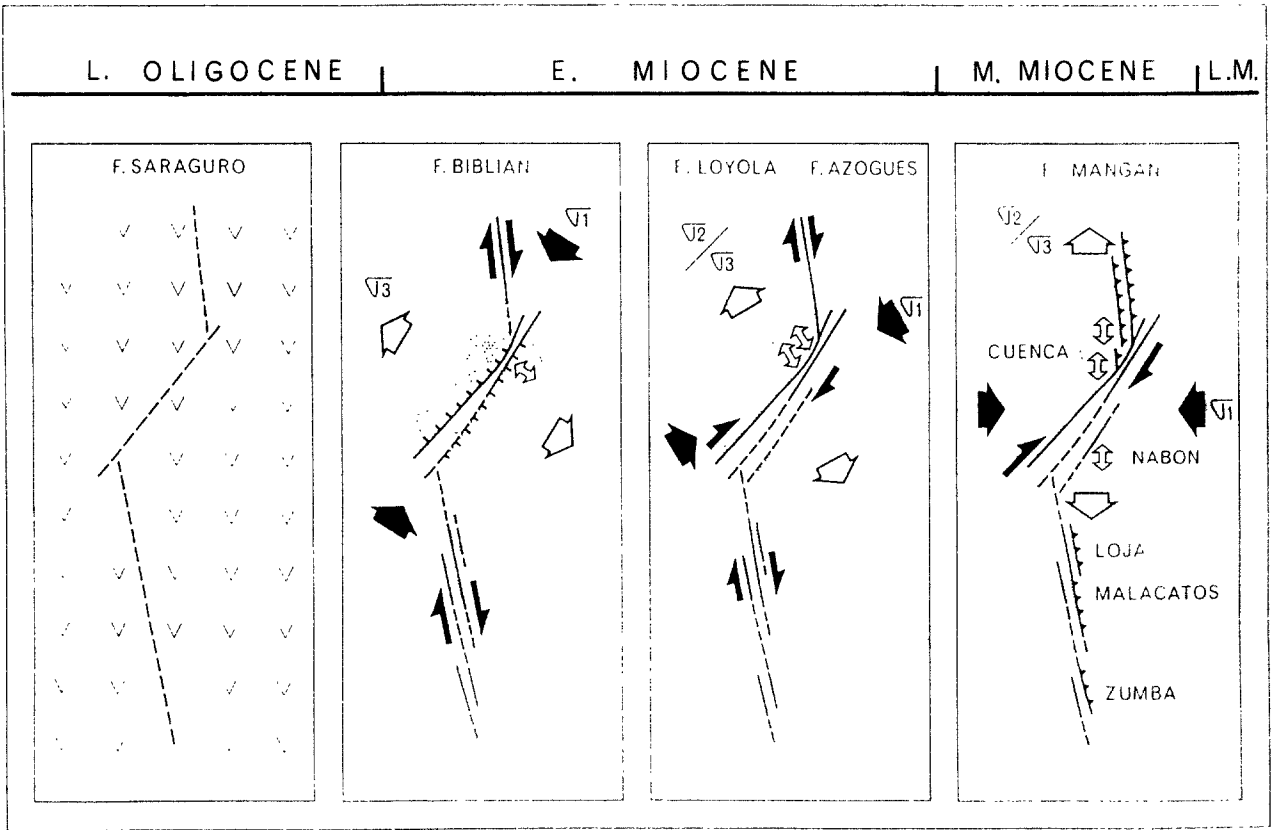


Fig. 5. Dynamic evolution of the Tertiary intermontane basins of southern Ecuador: large arrows indicate the principal stress directions; double open arrows indicate syn-sedimentary folded structures.

- 2) A middle sedimentary unit, with fluvatile and lacustrine deposits, also presents progressive unconformities.
 - 3) An upper volcanic unit, the Chalupas ash flow unit (Beate, 1985; Barberi *et al.*, 1988), rests on an erosion surface and transgresses the syn-sedimentary folded lower units.
 - 4) The final unit is a recent pyroclastic deposit that is called the Cangagua Formation.
- Until now, the second and third units were assigned to the presumed Pleistocene Latacunga For-

mation (Baldock, 1982). Two samples gave indistinguishable ages: FD86104, collected 7 km NE of Ambato on the right bank of the Rio Cutuchi from an outcrop shown on geologic maps (1978) as the Pisayambo Formation, yielded an age of 1.73 ± 0.35 Ma; FD87081, collected from the Latacunga Formation 2 km NE of Latacunga, yielded an age of 1.85 ± 0.19 M. The data indicate that these deposits are coeval and latest Pliocene in age.

Our field observations show that the first unit (previously assigned to the Pisayambo Formation)

and the second unit (Latacunga Formation, beneath the Chalupas flow) correspond in fact to the same volcano-sedimentary unit. We propose here to assign the first unit (former Pisayambo Formation in the Interandean Depression) and the second unit (lower part of the former Latacunga Formation) to the 'Latacunga Formation.' A sample (FD87110) collected on the east side of the Sagoatoa volcano, 10 km NNW of Ambato, and formerly assigned a Pliocene age, was dated at 1.40 ± 0.29 Ma (whole rock). Furthermore, Sagoatoa flows are interbedded in the Latacunga-Ambato basin-fill. These rocks are coeval with the Latacunga Formation and indicate a Pleistocene age for the upper part of this formation. Thus, the Latacunga Formation is late Pliocene to Pleistocene in age. The Chalupas ash-flow tuff lies on the eroded surface of the redefined folded Latacunga Formation. This observation is fully consistent with the age of 1.21 Ma established from the Chalupas caldera by Barberi *et al.* (1988).

NEOGENE TECTONIC IMPLICATIONS

The Tertiary to Quaternary tectonic evolution of the Ecuadorian Andes is related to the subduction of the Farallon-Nazca plates beneath the South American plate. The Neogene evolution of the Interandean basins of southern Ecuador was established by Noblet *et al.* (1988) and by Lavenu and Noblet (1989; 1990), who defined a transpressive then compressive tectonic continuum from latest Oligocene-early Miocene to Pliocene, lasting roughly 20 million years. Winter and Lavenu (1989a,b) and Winter (1990) provided a preliminary sketch of the Neogene tectonics.

In southern Ecuador, the lower Miocene formations unconformably overlie the Oligocene Saraguro Formation (Cuenca, Girón, and Nabon basins). The stress regime responsible for these deformations is still not well defined. The event apparently took place between 26.8 and 22 Ma, corresponding perhaps to the first stage of opening of the intermontane basins. It coincides with the breakup of the Farallon plate at about 26–25 Ma (late Oligocene). It is coeval, in Colombia, with the Miocene event at about 24 Ma (Duque-Caro, 1976), and in northern Peru, with the Quechua I phase of Noble *et al.* (1990). It corresponds, also, to a major tectonic event that affected all the Central Andes (Sébrier *et al.*, 1988; Sempéré *et al.*, 1990).

Interandean basins in southern Ecuador were affected by syn-sedimentary compressional deformation throughout the Miocene. The basins present the main characteristics of strike-slip basins, as defined by Nilsen and McLaughlin (1985). They are adjacent to a major intracontinental strike-slip fault system, with major faults oriented N–S (N170°E to N180°E) and NNE–SSW (N20°E to N40°E). Detailed structural analysis of folding and microtectonic faulting argues for continuous syn-sedimentary tec-

tonics. A geodynamic evolution of these basins has been proposed by Noblet *et al.* (1988) and Lavenu and Noblet (1989) (Fig. 5). The opening stage of the southern Ecuadorian basins is the result of a regional transpressive regime responsible for right-lateral movement along the N–S faults and for extensive movement along the NNE–SSW faults. The closing stage is also the result of a continuous regional transpressive then compressive regime which presents two principal directions of stress. During deposition of the Loyola and Azogues Formations, a NE/SW-trending shortening stress induced right-lateral displacement along all the main faults of the system. During deposition of the Mangan Formation, an E/W-trending shortening stress produced reverse movements on the N/S-trending faults and reverse-dextral movement on the NNE/SSW-trending faults.

The new dates obtained in southern Ecuador modify the timing of the events that Noblet *et al.* (1988) proposed. The basins could have begun to open during the late Oligocene (26–25 Ma). Closing could have started during the early Miocene (if the 16.3 Ma age is reliable), continuing until late Miocene (close to 8 Ma). Only the compressive stage is clearly observed elsewhere in the Andes:

- F3 compressive pulse in Peru (ca. 15–17 Ma; Sébrier *et al.*, 1988), and in Colombia (ca. 15 Ma; Duque-Caro, 1976).
- F4 pulse in Peru (ca. 10 Ma; Sébrier *et al.*, 1988), and in Colombia (ca. 10 Ma; Duque-Caro, 1976).
- F5 pulse in Peru (ca. 7 Ma; Sébrier *et al.*, 1988), and in Colombia (ca. 6–7 Ma; Van Houten, 1976).

In the Latacunga-Riobamba basin (central Ecuador), the latest Pliocene and the Quaternary Period are characterized by an E/W-trending syn-sedimentary compressional pulse that occurred before 1.21 Ma and was responsible for reverse movements on N/S-trending faults and associated N–S axis folds. Evidence of recent and active E/W-trending compressive tectonics has been recognized in central Ecuador along the Pallatanga fault (Winter and Lavenu, 1989a,b; Winter, 1990).

CONCLUSIONS

This study yields results that clarify the upper Paleogene, Neogene, and Quaternary stratigraphic, volcanic, and tectonic framework of southern and central Ecuador.

1. A long period of Oligocene volcanism is confirmed. In southern Ecuador, the age of the Saraguro Formation probably ranges from early to late Oligocene. In the Alausí area, the data demonstrate the existence of Oligocene rocks dating from 27 and 35 Ma, coeval with the Saraguro Formation but not belonging to the mapped Alausí Formation. Thus, volcanism occurred during the earliest Oligocene (≈ 35 Ma) and late Oligocene (29–27 Ma), with a possible hiatus between the two periods (*cf.* Soler,

1991). It is also noteworthy that, in some places, the mapped Macuchi Formation is mistaken for or overlain by Oligocene or Miocene rocks which were not distinguished in reconnaissance mapping. This applies to both the Alausí and Tarqui Formations.

2. From 25 to 26 Ma, two different geologic settings can be distinguished: (i) narrow intermontane sedimentary basins formed along major regional faults, providing relatively complete stratigraphic columns — in these basins, the products of volcanic episodes are interbedded with detrital deposits and radiometric age data can be compared with faunal contents; (ii) basin margins comprise mainly volcanic deposits, creating difficulties and sometimes confusion in mapping.

3. Miocene volcanism is confirmed to have been active from the southern to the central Ecuadorian Andes.

4. In southern Ecuador, no Pliocene ages have yet been obtained on volcanic rocks. These observations suggest that Plio-Pleistocene volcanism was not well developed in the south and that the Pleistocene Tarqui Formation was spatially more restricted than previously thought.

5. The Neogene tectonic framework of the southern and central Ecuadorian Andes is further defined and confirms a continuous transpressive then compressive syn-tectonic evolution during much of the Miocene Epoch in southern Ecuador, and compressive evolution from the late Pliocene to the Quaternary in central Ecuador.

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